CMPG 311 2026- Group 3

Real Estate Property Listing System

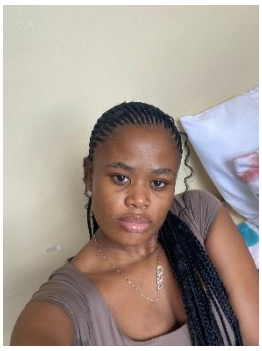
Phase 1

Database Initial Study

Module	CMPG 311 – Database Systems
Phase	Phase 1 – Database Initial Study
Topic	Real Estate Property Listing System
Submission Date	02 April 2026
Institution	North-West University (NWU)
Year	2026

1. Group Information

Ntshembo Mabasa
Group Leader
Role: Front-end and Database
Contact Number
0677527020
Email
47337354@mynwu.ac.za



Mmakoma Boshomane
Role: Front-end
Contact number
0797298054
Email
43723152@mynwu.ac.za

Khutso Choshane
Role: Back-end
Contact number
0824053762
Email
37774425@mynwu.ac.za

Shaun Molewa
Role: Front-end
Contact number
0728214859
Email
43628362@mynwu.ac.za

Piet Ribana
Role: Backend
Student No:
37140248
Contact Number
0685554065
Email
37140248@mynwu.ac
.za

Lebogang Motseki
Role: Front-end
Contact Number
068 477 3813
Email
37428136@mynwu.a
c.za

Madonsela Ronny
Role: Front-end and Database
Contact Number:
0792779959
Email:
39803384@mynwu.a
c.za

Mongale Kamogelo
Role: Back-end and Database
Contact Number
0780772170
Email
34663568@mynwu
.ac.za

Tebogo Morake
Role: Backend and Database
Contact Number
078 900 5970
Email
39510484@mynwu.a
c.za

Managa Wanani
Role: Database
Contact Number
0728794009
Email
44835507@mynwu.ac
.za

2. Company Situation Analysis

2.1 Company Overview

KeySpace is a mid-sized real estate agency operating in South Africa. Established to connect buyers, sellers, landlords, and tenants in the property market, the company provides end-to-end property listing and transactional services through a network of experienced agents. The company manages a diverse portfolio of residential, commercial, and rental properties.

2.2 Mission and Objectives

Mission Statement:

> *"To provide a seamless, transparent, and technology-driven real estate experience that connects clients with their potential home or next purchase through knowledgeable agents."*

Our project aims to improve and update the property search experience by bridging the gap between real estate agents and potential buyers. We will establish this by: creating a property listing website for real estate which consolidates property details in one place. Agents will be able to sign in on the website and submit property information such as location, price, size and type. We want to make a fast, relational data environment that puts everything in a structured digital catalogue instead of having to deal with chaotic file systems, hand-made spreadsheets and manual paperwork.

The system seeks to:

- Streamline property listings by agents.
- Enable clients to search for properties effortlessly, based on their preferences.
- Enhance interaction between agents and clients.
- Organise property information in a structured database.
- Facilitate the management, listing, searching, and monitoring of properties

Core Objectives:

- Maintain an up-to-date and comprehensive listing of available properties for sale and rent.
- Facilitate efficient and transparent interactions between agents and clients.
- Enable accurate property information based on location, price and property type.
- Track the full transaction lifecycle from listing to sale or lease agreement.
- Monitor agent performance and client satisfaction metrics.

2.3 Operations

The system is designed to support smooth workflows. Agents will log in securely to create and manage detailed property listings. Clients will browse these listings, apply filters to find suitable properties, and send inquiries directly to agents. An administrator will oversee user activities, monitor system performance, and maintain security

2.3.1 Property Listings Management

- Agents register new properties on behalf of sellers (for sale) or landlords (to let).
- Each listing includes property type, location, price or rental amount, size, features, and availability status.
- Listings are updated regularly to reflect sold, rented, or withdrawn properties.

2.3.2 Agent Operations

- Agents are assigned to specific regions or locations.
- Each agent manages a portfolio of property listings and client relationships.
- Agents schedule property viewings, negotiations, and follow-up communications.
- Agent performance is tracked through listing counts, viewings conducted, and deals closed.
- Management and updating of property listings

2.3.3 Client Management

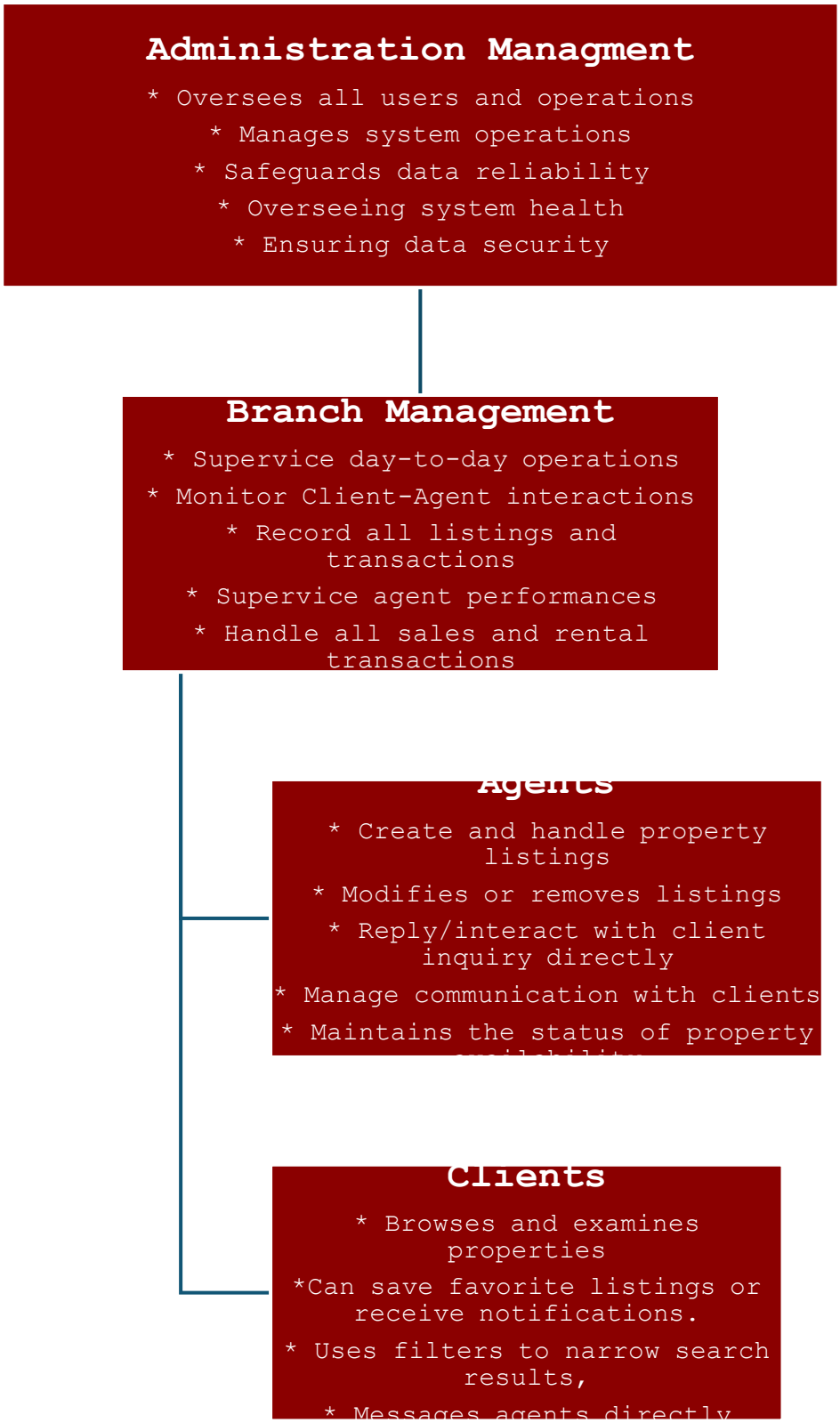
- Clients are categorised as buyers, sellers, tenants, or landlords.
- Client requirements and preferences are recorded to match them with suitable agents.
- The company maintains a history of all client interactions and transactions.
- User registration of clients to start using the platform.
- Direct contact with agents assigned to preferred properties.
- Management of their profiles like personal details and contact information.

2.3.4 Administration Management

- Managing tenant information.
- Recording and tracking of client-agent interaction.
- Monitor listings & agent performance.

2.4 Organisational Structure

A hierarchical organisational structure as detailed below in terms of access and accessing of information. **Role-based access** so that agents, managers, and administrative staff interact with appropriate data only.



3. Problems and Constraints

3.1 Identified Problems

Through analysis of **KeySpace** current operations, the following key problems have been identified:

Problem	Description
Lack of centralized property management	Property listings are maintained in spreadsheets and physical files, making it difficult to search, filter, or tracking availability. Duplicate listings and outdated information are common and scattered property information. Manual recording emphasizes errors in data entry. Data lacks a central repository.
Client-Agent Interaction Tracking	Agent-client interactions are cumbersome. There is no centralised system to track agent-client interactions, resulting in lost enquiries, missed viewings, and lack of follow-up on potential deals. Poor stakeholder communication & agents cannot track client inquiries effectively and no track of progress.
No Centralised Client Records	Client profiles, preferences, and transaction histories are stored informally, preventing agents from quickly accessing client information or preferred property matches.
Scalability	Supporting a growing number of users, properties, and property images. Data risks loss or redundancy. Data is stored in fragmented locations. The system must accommodate future growth in the number of listings, agents, and branches without requiring a complete redesign.
Security	Protecting sensitive user data and preventing unauthorized access or fraudulent listings. Abiding by the South African POPI Act for all documents submitted by having good storage and handling.

3.2 Constraints

Constraints	Description
Access	Certain members of the team/company can access certain parts of the system/database. Admin oversees operations, agents can't see transactions, etc.
Data Migration	Existing data held in spreadsheets and paper records must be validated and migrated, introducing risks of data loss and inconsistency. Ensuring data integrity and consistency.
Internet dependency	The system requires a stable internet connection for access.
Data Privacy/Security	Client personal information and financial data must be handled in compliance with the Protection of Personal Information Act (POPIA). Sensitive user information must be protected, and passwords must be securely hashed. Sensitive user information must be protected, and passwords must be securely hashed.
Scalability	The system must accommodate handle multiple users accessing and using it at the same time.
Regulatory Compliance	The system must comply with legal and ethical standards related to real estate advertising and user data protection. Compliance with regulations
Property Listing Rules	Only verified agents or authorized users should be able to post listings
Budget	Monetary limits for development resources.

4. Database System Specifications

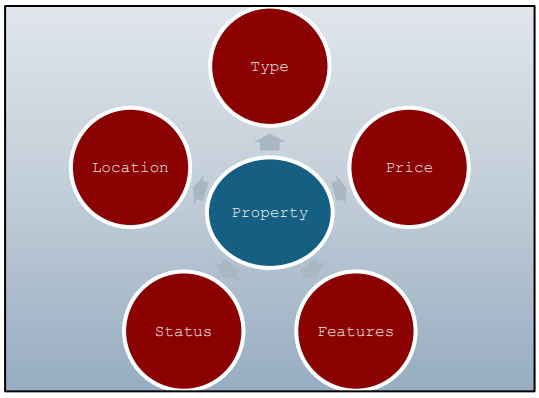
4.1 Objectives of the Database System

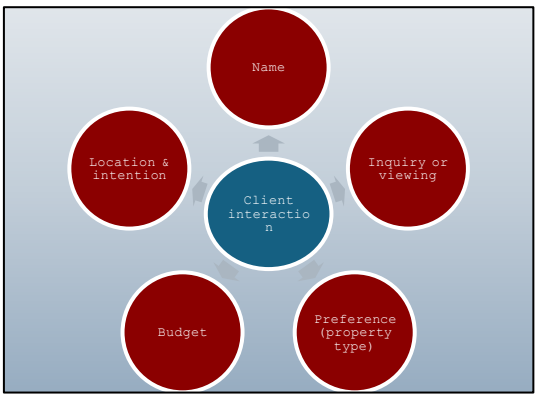
The database and system are designed to securely store detailed property, user, and inquiry information/history. They aim to support quick data retrieval, maintain data accuracy, and establish clear relationships between users, agents, and properties. The database system is designed to address the identified problems through the following objectives:

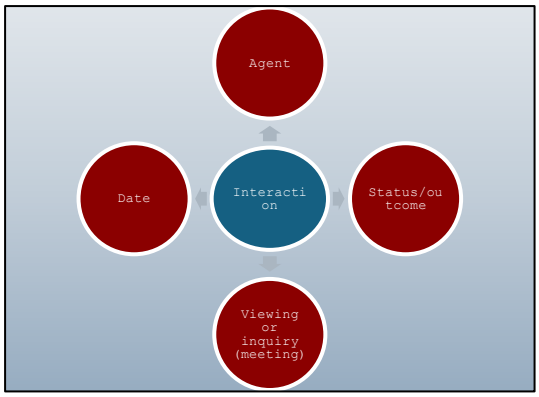
- **Store sale and rental listings:** Provide a centralised, searchable repository for all property listings with accurate and current information stored securely.
- **Record and track:** All agent-client interactions including enquiries, viewings, outcomes, and follow-ups. To improve communication between property seekers and agents. Manage users and agents and overall track tenants.
- **Maintain client profiles:** The secure storage of all registrations of user details including personal details, preferences, and transaction/interaction history. As users browse, search, save their favourite listings **or receive notifications** and inquire about properties.
- **Capture and monitor:** To track the full lifecycle of each property transaction from initial listing to final sale or leasing.
- **Generate reports:** Enable management to generate reports on agent performance, listing volumes, transaction values, and market trends.
- **Enforce data integrity through constraints, referential integrity, and validation rules.**
- Support **role-based access** so that agents, managers, and administrative staff interact with appropriate data only.

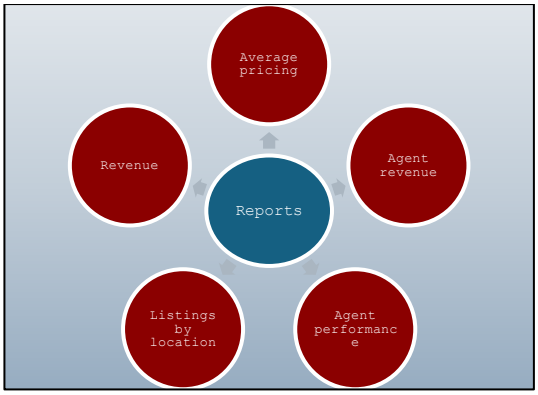
4.2 Information Required from the Database

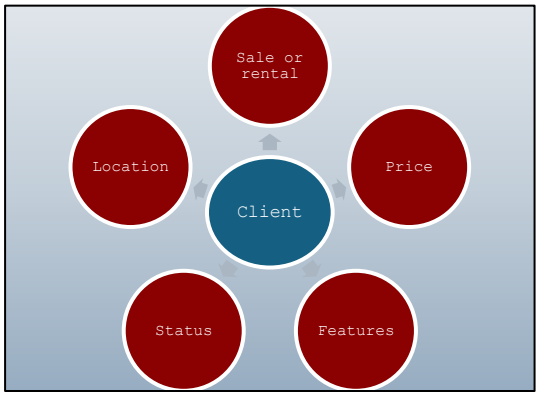
The following information outputs are required from the database to support business operations:











4.3 Scope

The database system will cover the following areas:

- **Property listings:** All sales and rental listings managed by the company of all categories, including residential (houses, apartments, townhouses) and commercial properties.
- **Registrations/ user accounts:** All registered agents and their branch and regional assignments. All clients registered with the company of all categories, including buyers, sellers, tenants, just inquiries and landlords.
- **Search functionality:** All property sale and rental transactions processed through Prime Properties.
- **Reports and analysis:** Performance and reporting data for management decision-making.
- **Communication tools:** The system will support direct messaging between clients and agents, user inquiries and viewing requests.

The following are explicitly OUT OF SCOPE for this system:

- Legal conveyancing processes and external attorney systems.
- Bond transfers and bond approval processing (handled by external financial institutions).
- Clients/Buyers/renters can only browse, save, and inquire about properties.
- Accounting and payroll systems for agent remuneration. Including online transactions.

4.4 Boundaries

The system boundaries define what the database will and will not interact with:

WITHIN BOUNDARY (In Scope)	OUTSIDE BOUNDARY (Out of Scope)
✓ Property listing data and status management	✗ External property portals and their data synchronisation
✓ Agent profiles and locations	✗ Agent payroll and commission payment processing
✓ Client registration and preference management	✗ Legal documentation and conveyancing processes (property ownership transfers)
✓ Agent-client interaction logging	✗ External bond and bond origination systems
✓ Sale and rental transaction records	✗ Municipal rates and utilities billing systems
✓ Management performance reporting	✗ Third-party property valuation service integrations

END